

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

PRN / Roll No: _____

Division: _____

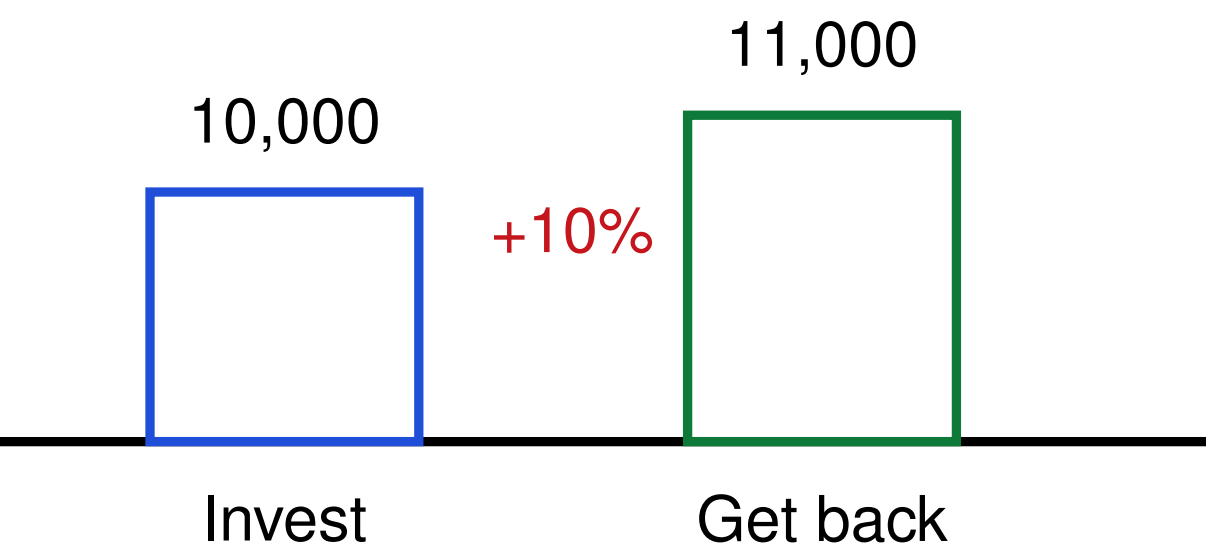
UNIT V: BASICS OF INVESTMENT

1. Investment Return

The gain on an investment, as a % of the money invested.

$$\text{Return} = \frac{\text{Gain}}{\text{Invested}} \times 100$$

Eg: Invest Rs 10,000, get back Rs 11,000.
Return = $1,000 \div 10,000 = 10\%$

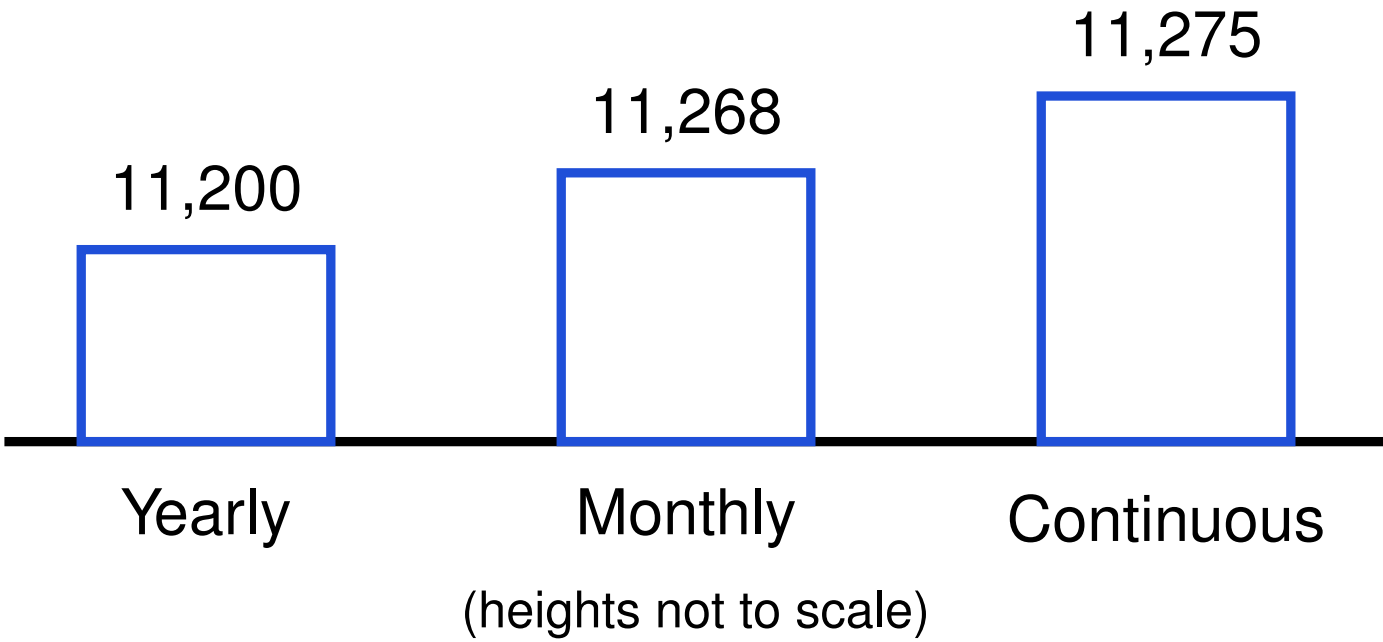


2. Compounding Frequency

The more often interest is added in a year, the more money you get.

$$A = P \left(1 + \frac{r}{m} \right)^{mt}$$

Eg: Rs 10,000 at 12% for 1 year

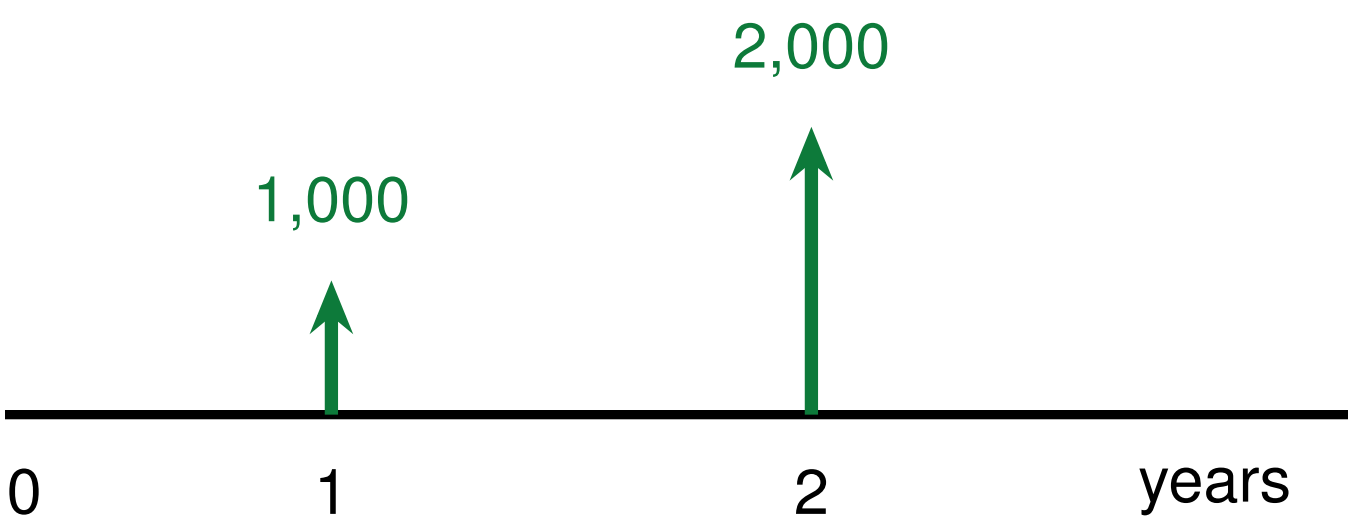


3. Uneven Cash Flows

Money received is different every year. Bring each amount back to today.

$$PV = \frac{CF_1}{(1+i)} + \frac{CF_2}{(1+i)^2}$$

Eg: Rs 1,000 (year 1) and Rs 2,000 (year 2), $i = 10\%$
 $PV = 909 + 1,653 = \text{Rs } 2,562$

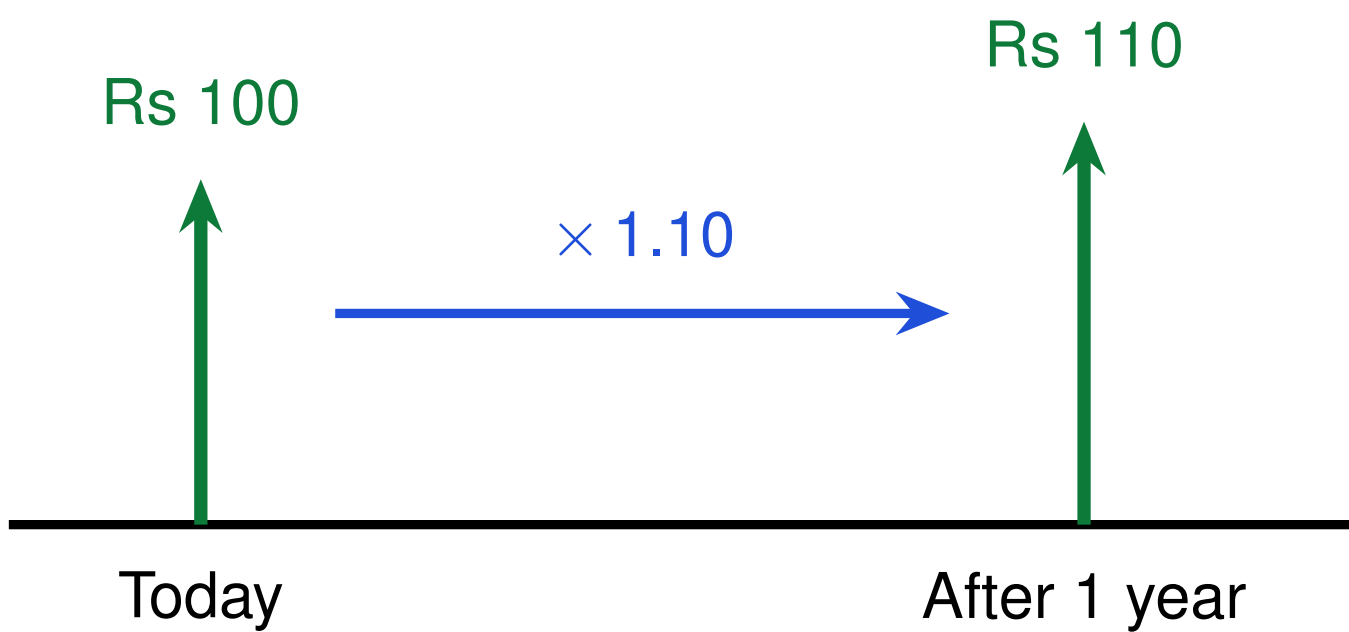


4. Economic Equivalence

Different amounts at different times can have the same value.

$$F = P(1+i)^n$$

Eg: at 10%, Rs 100 today = Rs 110 after 1 year



UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk

Chances are known. Choose the best average (EMV).

$$\text{EMV} = \sum (\text{chance} \times \text{payoff})$$

Eg: 50%: Rs 100, 50%: Rs 40 $\Rightarrow$ EMV = Rs 70



2. Decision under Uncertainty

Chances are unknown. Use a rule:

- **Maximax**: best of the best
- **Maximin**: best of the worst

Payoff	Boom	Slump
Stock	80	-20
Bond	40	20

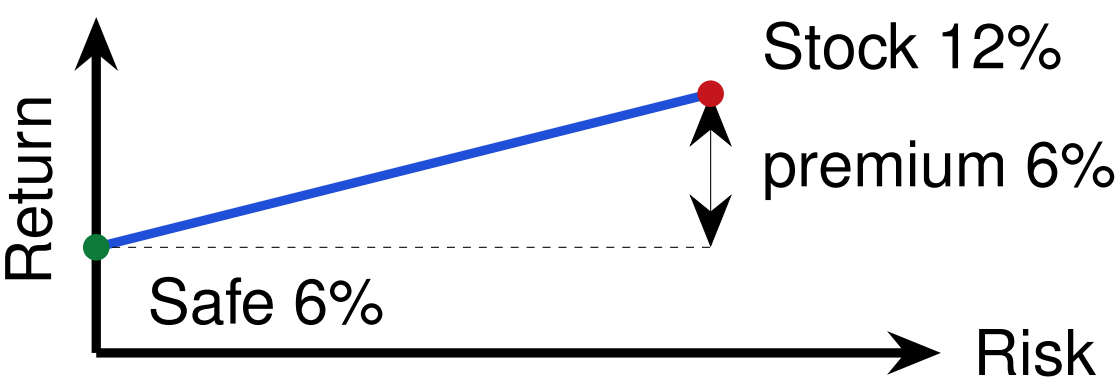
Maximax $\rightarrow$ Stock Maximin $\rightarrow$ Bond

3. Risk Premium

The extra return you ask for when taking risk.

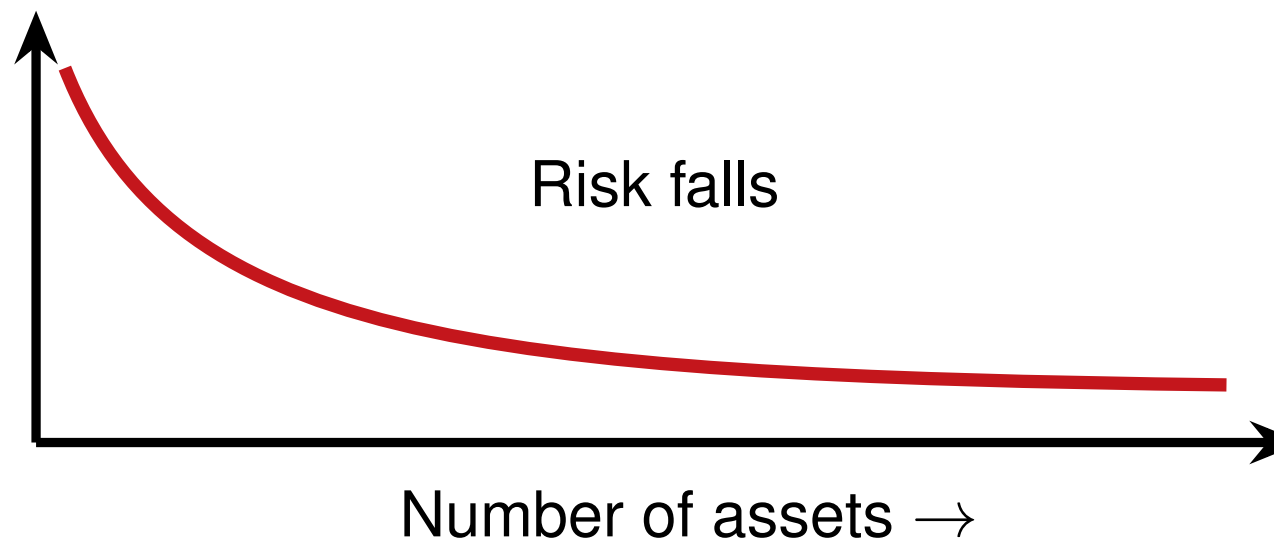
$$\text{Premium} = \text{Expected} - \text{Safe rate}$$

Eg: $12\% - 6\% = 6\%$



4. Portfolio Diversification

Don't put all eggs in one basket: more different assets means less risk.

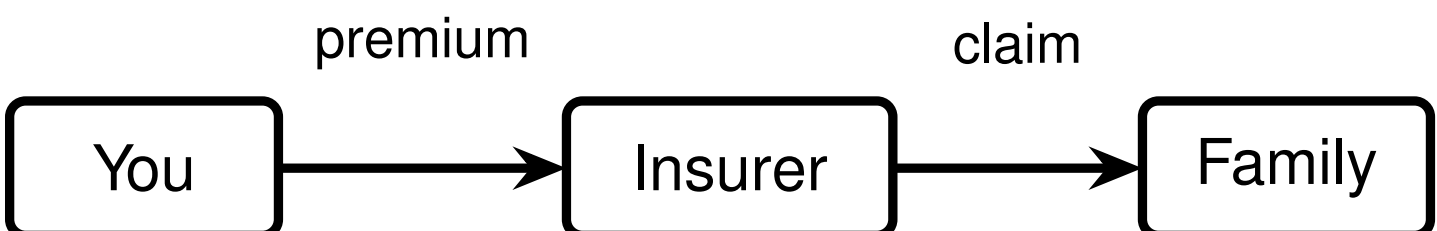


5. Life Insurance

You pay a premium. The insurer pays the sum assured if you die.

$$\text{Premium} \approx \text{Sum assured} \times P(\text{death})$$

Eg: $\text{Rs } 1,00,000 \times 0.01 = \text{Rs } 1,000$



6. Endowment

Pays if you die in the term OR survive to the end.

$$\text{Endowment} = \text{Insurance} + \text{Savings}$$



Higher risk $\Rightarrow$ higher expected return